



COZY HANDS



A PROJECT REPORT

Submitted by

ABITHA K R	2303811710622003
AKALYA B	2303811710622008
GAYATHRI K K	2303811710622026

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(AUTONOMOUS)**

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**K. RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(AUTONOMOUS)**

SAMAYAPURAM – 621112

BONAFIDE CERTIFICATE

The work embodied in the present project report entitled “**COZY HANDS**” has been carried out by the students Abitha K R, Akalya B, Gayathri K K. The work reported herein is original and we declare that the project is their own work, except where specifically acknowledged, and has not been copied from other sources or been previously submitted for assessment.

Date of Viva-voce:

Dr. A. PUNITHA, M.E.,Ph.D.

SUPERVISOR

Professor

Department of Electronics and

Communication Engineering

K Ramakrishnan College of

Technology (Autonomous)

Samayapuram-621112

Dr. S. SYEDAKBAR, M.E.,Ph.D.

HEAD OF THE DEPARTMENT

Assistant Professor

Department of Electronics and

Communication Engineering

K Ramakrishnan College of

Technology (Autonomous)

Samayapuram-621112

INTERNAL EXAMINER

EXTERNAL EXAMINER

ABSTRACT

The Cozy Hands project presents a reusable and battery-free hand-warming glove designed to provide thermal comfort in cold environments. The system uses a supersaturated sodium acetate solution, which generates heat through an exothermic crystallization process when activated by a snap button. This heat is safely transferred through a sealed pouch integrated inside the glove, ensuring effective and uniform warming of the hands. The glove is made using flexible and durable materials to provide comfort, proper fitting, and efficient heat distribution. Unlike conventional heating methods, this system does not require electricity or batteries, making it lightweight, portable, and reliable for everyday use. One of the major advantages of the system is its reusability, as the sodium acetate solution can be reheated and restored to its original state for multiple uses. The proposed design is cost-effective, eco-friendly, and safe, making it a suitable alternative to disposable and battery-operated hand warmers. Overall, the project offers a practical and sustainable solution for maintaining hand warmth in cold conditions.

Keywords

Sodium acetate, hand warmer, exothermic reaction, crystallization, phase change material, thermal energy.

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LIST OF ABBREVIATIONS

ET	-	Exothermic Transformation
HT	-	Heat Transfer
PCM	-	Phase Change Material
PPE	-	Personal Protective Equipment
SC	-	Supercooled Condition
TC	-	Thermal Comfort
TE	-	Thermal Energy
TM	-	Temperature Measurement

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

Maintaining adequate hand warmth in cold environmental conditions is essential for comfort, health, and effective performance of daily activities. Exposure to low temperatures can lead to reduced blood circulation, numbness, stiffness, and decreased manual efficiency. In severe cases, prolonged exposure may result in cold stress or frostbite. Conventional gloves provide passive insulation by trapping body heat; however, they may not offer sufficient protection in extremely cold climates. Therefore, an active and reusable heating mechanism integrated within wearable accessories becomes necessary to enhance thermal comfort and safety.

The proposed system incorporates a reusable heat-generating mechanism within gloves using a supersaturated sodium acetate solution enclosed in a secure zip-lock or zipper pouch. The heat generation process is based on an exothermic crystallisation reaction. When the embedded metal snap disc is pressed, it initiates nucleation, causing rapid crystallisation of the sodium acetate solution. During this phase change from liquid to solid, latent heat is released, thereby increasing the temperature within the glove and providing warmth to the user's hands. The reaction is safe, controlled, and capable of maintaining heat for a considerable duration. After use, the crystallised solution can be restored to its liquid state by heating it in hot water, allowing the system to be reused multiple times. This reusability makes the design environmentally friendly and cost-effective compared to disposable heating alternatives.

The gloves are fabricated using nylon spandex material, selected for its durability, elasticity, flexibility, and comfort. Nylon contributes strength and resistance to wear, while spandex ensures a snug fit and ease of movement. The integration of the heating pouch within the glove structure ensures efficient heat

transfer while maintaining user safety and convenience. The zip-lock or zipper mechanism facilitates easy removal and reheating of the heating element without damaging the glove. By combining principles of thermodynamics, phase change reactions, and textile engineering, the design offers a practical and sustainable solution for improving hand comfort in cold conditions. Such a system can be effectively utilised in outdoor activities, cold storage environments, winter travel, and other low-temperature applications where maintaining hand warmth is essential.

1.2 BACKGROUND

Exposure to cold environments can significantly affect human comfort and performance, especially in the hands. Hands are one of the most sensitive parts of the body to temperature changes because they have a high surface area and are often directly exposed to cold air. In low-temperature conditions, blood circulation to the fingers reduces to conserve body heat, leading to numbness, stiffness, and decreased grip strength. Prolonged exposure to extreme cold can even cause cold stress or frostbite. Therefore, maintaining proper warmth in the hands is essential for ensuring comfort, safety, and efficiency in daily activities, particularly for individuals working outdoors, travelling, or living in colder regions.

Traditional methods of keeping hands warm mainly rely on woollen gloves or layered clothing, which provide insulation but do not actively generate heat. Electrically heated gloves are another option, but they depend on batteries, which increase weight, cost, and maintenance requirements. Battery-operated systems may also pose safety risks and require frequent charging. Hence, there is a growing need for a portable, safe, reusable, and cost-effective solution that can provide warmth without relying on electrical energy.

Sodium acetate-based reusable hand warmers offer an efficient alternative. Sodium acetate has the ability to remain in a supercooled liquid state until triggered. When the metal snap button inside the pouch is pressed, it initiates a crystallisation process. This crystallisation is an exothermic reaction, meaning it releases heat to the surroundings. As the liquid converts into solid crystals, heat is generated instantly and can last for 20 to 40 minutes. After use, the solidified crystals can be converted back

into liquid form by heating the pouch in boiling water, making the hand warmer reusable and environmentally friendly.

In this project, the sodium acetate hand warmer is integrated directly into gloves to improve convenience and heat distribution. Instead of carrying a separate heating pack, the warmer is placed inside a zip-lock or zipper pouch within the glove. This design ensures safety, prevents leakage, and allows easy removal for reheating. Integrating the heat source into the glove allows better contact with the palm and fingers, providing uniform warmth where it is most needed.

The glove is made using nylon spandex material, which combines durability, flexibility, and comfort. Nylon provides strength and resistance to wear and tear, while spandex ensures elasticity and a snug fit. The material is lightweight, breathable, and suitable for daily use, allowing free hand movement while maintaining effective heat transfer. Overall, this project aims to design a reusable, safe, economical, and eco-friendly hand-warming glove system that reduces cold stress and enhances user comfort in low-temperature environments.

1.3 OBJECTIVE OF THE PROJECT

The primary objective of this project is to design and develop a reusable, safe, and efficient hand-warming system integrated into gloves using sodium acetate as the heat-producing material. The project aims to utilise the exothermic crystallisation reaction of sodium acetate, which is activated by a snap metal button, to generate instant heat when exposed to cold environments. The heating element is securely enclosed within a durable zip-lock or zipper pouch and embedded inside gloves made of nylon spandex material to ensure flexibility, comfort, and proper fitting. This design focuses on providing uniform heat distribution across the hands to reduce cold stress, prevent numbness, and improve overall hand comfort during low-temperature conditions.

Additionally, the project aims to create a cost-effective, portable, and eco-friendly alternative to disposable hand warmers by promoting reusability. The system is designed to be easily recharged by reheating the sodium acetate solution after crystallisation, making it sustainable for long-term use. Another objective is to ensure user safety by preventing leakage and maintaining proper insulation within the glove structure. Overall, the project seeks to enhance thermal protection for individuals

working or travelling in cold climates while maintaining comfort, durability, and practicality.

1.4 NOVELTY OF THE PROJECT

The novelty lies in the development of a reusable hand-warming glove system that integrates a sodium acetate-based heat pack directly into a nylon-spandex glove through a secure zip-lock pouch design. Unlike conventional hand warmers that are kept separately inside gloves, this design ensures proper placement and uniform heat distribution. The integrated structure prevents movement of the heat pack and enhances overall user comfort during use.

A supersaturated sodium acetate solution is used to generate heat through a controlled exothermic crystallisation reaction when activated by a snap metal disc. The activation is completely user-controlled, allowing heat production only when required. Since no batteries or external power supply are needed, the system is energy-efficient, portable, and safe for use in cold environments. The selection of nylon-spandex material adds further innovation to the design. This fabric provides flexibility, stretchability, and a snug fit, which helps retain heat effectively while maintaining comfort. Its lightweight and breathable nature allows extended wear without discomfort, making it suitable for daily use in low-temperature conditions. In addition to these features, the design introduces an improved ergonomic integration of the heat pack within the glove, ensuring that the warmth is distributed evenly across critical areas such as the fingers and palm. This targeted heating approach enhances thermal efficiency and provides better comfort compared to traditional localized heating methods.

The use of Sodium Acetate in a sealed zip-lock pouch further contributes to the system's reliability by preventing leakage and ensuring safe operation over multiple usage cycles. The pouch design also allows easy removal and replacement if required, adding to the modularity of the system. Another novel aspect of the project is its focus on sustainability and reusability. Unlike disposable chemical warmers, this system can be regenerated repeatedly with minimal environmental impact.

CHAPTER 2

LITERATURE SURVEY

Cold exposure and its physiological impact on the human body have been extensively discussed in biomedical literature. J. Castellani and T. Young,(2025) in their paper titled “Human Physiological Responses to Cold Exposure: Acute Responses and Acclimatisation to Prolonged Exposure” [1], explain that exposure to low temperatures causes rapid peripheral vasoconstriction, reducing blood flow to extremities such as fingers and hands. This reduction in circulation leads to numbness, stiffness, and loss of fine motor skills. Their work emphasises that maintaining hand warmth is essential not only for comfort but also for preserving manual performance and preventing cold-related injuries. Supporting this, D. A. Daanen(2024) in “Finger Cold-Induced Vasodilation: A Review” [2] discusses how repeated cold stress affects finger blood flow regulation and significantly reduces hand functionality. These physiological findings clearly justify the need for effective hand-warming technologies.

Traditional gloves have long been used as a primary method of thermal protection. Y. Li,(2024) in “The Science of Clothing Comfort” [3], explains that thermal insulation in garments works by trapping air within textile fibres, thereby reducing heat transfer through convection and conduction. Materials such as wool, fleece, and synthetic polymers provide passive insulation that slows down heat loss. However, Li notes that passive systems cannot generate heat; they only conserve the body’s existing warmth. Similarly, S. Havenith(2024) in “Heat Balance When Wearing Protective Clothing” [4] highlights that insulation effectiveness decreases significantly under extreme cold conditions. Once hand temperature drops below a certain level, passive insulation alone becomes insufficient to restore thermal comfort. These limitations indicate the need for active or semi-active heating solutions.

To address the shortcomings of passive systems, electrically heated gloves were developed. S. Park, J. Lee, and H. Kim,(2023) in “Development of Electrically Heated

Gloves for Cold Environment Applications” [5], describe gloves integrated with resistive heating wires powered by rechargeable lithium-ion batteries. Their study demonstrates controlled heating within the comfort range of 35°C to 45°C. However, the researchers also report challenges such as increased glove weight, battery dependency, limited operating time, and reduced battery efficiency in low temperatures. Furthermore, M. Stoppa and A. Chiolerio (2023) in “Wearable Electronics and Smart Textiles: A Critical Review” [6] discuss broader issues associated with wearable heating electronics, including circuit complexity, potential safety risks, and higher production costs. While electrically heated gloves offer temperature control, their reliance on power sources limits sustainability and affordability.

Disposable chemical hand warmers have also gained popularity. K. Yamashita(2022) in “Performance Evaluation of Iron Oxidation-Based Disposable Warmers” [7] explains that these warmers function through the exothermic oxidation of iron powder when exposed to air. The reaction provides heat for several hours, making them convenient for short-term use. However, Yamashita notes that these products are single-use and contribute to environmental waste. Additionally, temperature control is not adjustable once the reaction begins. These drawbacks encourage exploration of reusable thermal systems.

Phase Change Materials have emerged as promising alternatives for thermal energy storage. A. Sharma, V. Tyagi, C. Chen, and D. Buddhi,(2022) in their review “Review on Thermal Energy Storage with Phase Change Materials and Applications” [8], describe how PCMs absorb and release latent heat during phase transitions. Sodium acetate trihydrate is particularly highlighted due to its high latent heat storage capacity, chemical stability, and non-toxic nature. The study confirms that sodium acetate can remain in a supercooled liquid state and release stored heat rapidly upon crystallisation. Similarly, F. Agyenim, N. Hewitt, P. Eames, and M. Smyth(2021) in “A Review of Materials, Heat Transfer and Phase Change Problem Formulation for Latent Heat Thermal Energy Storage Systems” [9] emphasise the suitability of PCMs for compact and reusable thermal applications.

R. Lane (2021) in “Reusable Sodium Acetate Heat Packs for Therapeutic for

Applications” [10] explains that a metallic snap disc acts as a nucleation trigger when flexed, initiating crystallisation in the supercooled solution. This process releases heat up to approximately 54°C. The author highlights key advantages such as reusability through boiling, absence of toxic byproducts, and independence from electrical power. Additionally, J. D. MacDonald (2021) in “Supercooling and Crystallisation in Sodium Acetate Solutions” [11] studies the thermodynamic properties of sodium acetate and confirms stable and repeatable heat release characteristics, making it suitable for portable thermal systems.

Research on wearable system integration has further explored embedding functional components into textiles. L. Van Langenhove (2021) in “Smart Textiles for Healthcare Applications” [12] emphasises the importance of ergonomic design, flexibility, and user safety in wearable thermal products. While most wearable heating technologies rely on electronic elements, limited research has focused on integrating chemical-based heating systems into gloves in a structured and ergonomic manner.

From the reviewed literature, several important observations emerge. Cold exposure significantly reduces hand circulation and performance [1], [2]. Passive insulation methods cannot actively restore lost heat [3], [4]. Electrically heated gloves provide controlled warmth but introduce dependency on batteries and higher costs [5], [6]. Disposable chemical warmers generate environmental waste [7]. Sodium acetate-based phase change systems offer a safe, reusable, and power-free alternative [8], [10], [11].

CHAPTER 3

PROPOSED METHOD

3.1 BLOCK DIAGRAM

The proposed hand warmer glove system is designed as a simple, battery-free thermal device that operates based on a chemical heat generation process. The system can be divided into several functional blocks that work sequentially to produce and distribute heat efficiently. The first block is the User Activation Block, where the user manually presses the snap button embedded inside the sodium acetate pack. This step is crucial as it initiates the entire process without requiring any external power source or electronic control. The snap button acts as a mechanical trigger, making the system easy to operate for all users.

The next stage is the Trigger Mechanism Block, which consists of a stainless-steel snap disc. When the disc is flexed, it creates nucleation points within the supercooled sodium acetate solution. This disturbance breaks the stable liquid state and initiates crystallization. This block ensures that the reaction starts instantly and reliably whenever required. Following this, the system enters the Chemical Reaction Block, where the core process occurs. Sodium acetate, which is initially in a supercooled liquid state, undergoes an exothermic crystallization reaction. During this phase change from liquid to solid, latent heat energy stored in the solution is released. This reaction generates a significant amount of heat without any combustion or electrical input, making it safe and efficient.

The generated heat is then passed to the Heat Transfer Block. In this stage, the heat moves from the sodium acetate pack to the surrounding glove layers through conduction. The design ensures minimal heat loss and maximum transfer efficiency so that the user experiences effective warming. Next is the Heat Distribution Block, which includes the inner lining and insulation layers of the glove. These layers help in spreading the heat uniformly across the hand, preventing localized overheating and ensuring user comfort. Finally, the system reaches the Output Block, where the user

experiences a warming effect. The generated heat improves hand comfort, maintains proper blood circulation, and reduces the effects of cold exposure such as numbness and stiffness. After the heat energy is exhausted, the system moves into the Regeneration Block. In this stage, the solidified sodium acetate is reheated in hot water until it returns to its liquid state. Once cooled, it becomes ready for reuse. This feature makes the system environmentally friendly and cost-effective.

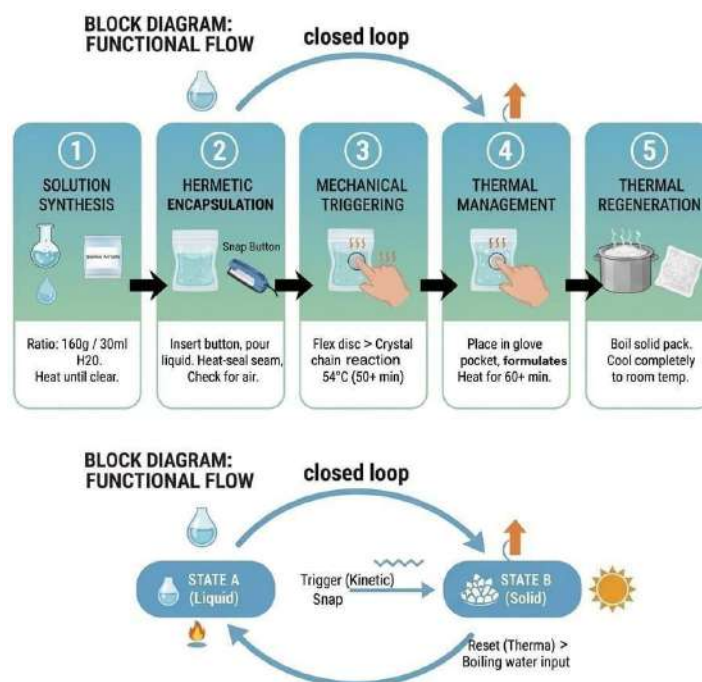


Figure 3.1 Block Diagram of Cozy Hands

3.2 COMPONENTS USED

- Sodium Acetate Solution
- Snap Metal Button
- Leak-Proof Plastic Pouch
- Insulated Gloves
- Inner Lining Fabric
- Outer Protective Layer

3.3 WORKING PRINCIPLE AND IMPLEMENTATION

The proposed hand warmer glove operates based on the principle of an exothermic crystallization reaction using sodium acetate. Initially, the sodium acetate solution is maintained in a supercooled liquid state, where it remains stable even below its normal crystallization temperature. This stored state contains latent heat energy. When the user presses the snap button (metal disc) inside the pack, it introduces a disturbance that acts as a nucleation point. This triggers the crystallization of sodium acetate, causing it to rapidly change from liquid to solid form. During this phase transition, the stored latent heat is released in the form of thermal energy. The generated heat raises the temperature of the pack to a comfortable range, typically around 50–55°C, which is suitable for warming the hands without causing harm. The heat is then transferred to the surrounding glove material, providing warmth and improving blood circulation. This entire process occurs without the use of batteries or electrical components, making the system safe, simple, and energy-efficient. After the heat is dissipated, the sodium acetate solidifies completely. To reuse the system, the pack is placed in hot water until the crystals dissolve back into a liquid state, restoring the supercooled condition for the next cycle.

The implementation of the system involves integrating the sodium acetate heat pack into a wearable glove structure. A sealed, leak-proof plastic pouch containing the sodium acetate solution and snap disc is placed inside a specially designed pocket within the glove. The placement is typically at the palm or back of the hand to ensure effective heat distribution. The glove is constructed with multiple layers. The inner layer provides comfort and allows uniform heat transfer to the skin. The middle layer securely holds the heat pack in position, preventing movement during usage. The outer layer is made of insulating material to minimize heat loss to the environment and maintain warmth for a longer duration.

The snap button is positioned in such a way that it can be easily pressed by the user without removing the glove. This ensures quick activation even in cold conditions. The design avoids the use of wires, batteries, or electronic components, reducing complexity and enhancing reliability.

Safety is ensured by using non-toxic sodium acetate and maintaining a controlled temperature range. The system is reusable, cost-effective, and environmentally friendly, making it suitable for everyday use in cold environments. In addition to its basic functionality, the design can be further enhanced by optimizing the size and shape of the heat pack to ensure uniform heat distribution across the entire hand. Flexible and ergonomically designed pouches can improve comfort and allow better adaptability to different hand sizes. The selection of insulating materials for the outer layer plays a crucial role in extending the heat retention time and improving overall efficiency.

Furthermore, safety considerations are incorporated by ensuring that the temperature remains within a controlled range and that the pouch material is durable enough to withstand repeated heating cycles without leakage. The use of Sodium Acetate also ensures that the system remains non-toxic and environmentally safe for regular use.

3.4 SODIUM ACETATE SOLUTION

Sodium acetate solution serves as the fundamental heat-generating medium in the proposed hand warmer glove system. It is prepared in a supersaturated or supercooled liquid state, which is a metastable condition where the solution remains in liquid form even below its normal freezing point. In this state, the solution stores a significant amount of latent heat energy. When triggered, the solution undergoes rapid crystallization, releasing the stored energy in the form of heat through an exothermic reaction. This heat is sufficient to raise the temperature of the pack to a comfortable and safe level for human use. One of the key advantages of sodium acetate is its chemical stability and non-toxic nature, making it suitable for direct integration into wearable systems. Furthermore, the material supports multiple reuse cycles. After crystallisation, the solidified sodium acetate can be reheated in boiling water to dissolve the crystals and restore it to its original liquid state. Upon cooling, it regains its supercooled condition, ready for reuse. This reversible process makes the system highly sustainable and cost-efficient compared to disposable heating solutions.



Figure 3.2 Sodium Acetate

3.5 SNAP METAL BUTTON

The snap metal button plays a critical role as the activation mechanism in the system. It is typically a small, flexible stainless-steel disc placed inside the sodium acetate pouch. The disc is designed to produce a snapping action when pressed, which generates a localised disturbance in the solution. This disturbance introduces nucleation sites, breaking the metastable equilibrium of the supercooled liquid and initiating the crystallisation process. The design of the snap disc ensures reliable and instantaneous activation, allowing the user to generate heat whenever required. Unlike electronic triggers, this mechanical method is simple, durable, and does not require any external power source. The snap button is capable of withstanding repeated use without significant wear, making it highly suitable for reusable applications. Its placement within the pouch is carefully designed to ensure easy accessibility while maintaining safety and efficiency.



Figure 3.3 Snap Metal Button

3.6 LEAK-PROOF PLASTIC POUCH

The leak-proof plastic pouch acts as a container that holds the sodium acetate solution and the snap metal button securely. It is made from high-quality, flexible polymer materials that are resistant to heat and mechanical stress. The pouch is

hermetically sealed to prevent leakage, ensuring safe handling and long-term durability. Its flexible nature allows it to adapt to the shape of the hand, which improves contact and enhances heat transfer efficiency. The pouch also protects the solution from external contamination and prevents direct contact between the chemical and the user's skin. Additionally, it is designed to withstand repeated cycles of heating and cooling during the regeneration process. The quality and design of the pouch are critical, as they directly influence the safety, durability, and performance of the entire system.



Figure 3.4 Leak-Proof Plastic Pouch

3.7 INSULATED GLOVES

The insulated gloves serve as the main wearable structure of the system, providing both support and thermal insulation. These gloves are typically made from materials such as wool, fleece, or synthetic thermal fabrics, which have low thermal conductivity. This helps in reducing heat loss to the surrounding environment and ensures that the heat generated by the sodium acetate pack is retained within the glove. The gloves are designed to comfortably fit the user's hand while accommodating the heat pack within a designated compartment. They also protect the user from external cold air, wind, and moisture. The insulation provided by the gloves enhances the effectiveness of the heating system by maintaining a stable temperature for a longer duration. Additionally, the gloves are lightweight, flexible, and suitable for daily use, making them practical for various applications such as outdoor activities and cold-weather conditions.



Figure 3.5 Insulated Gloves

3.8 INNER LINING FABRIC

The inner lining fabric is an essential component that ensures comfort and proper heat distribution. It is made from soft, breathable, and skin-friendly materials such as cotton or fleece. This layer comes in direct contact with the user's hand and plays a crucial role in evenly distributing the heat generated by the sodium acetate pack. By preventing direct concentration of heat in a single area, it avoids discomfort and ensures a uniform warming effect. The inner lining also absorbs moisture, helping to keep the hand dry during prolonged usage. Its breathable nature improves overall comfort and prevents irritation, making the glove suitable for extended wear. The selection of appropriate inner lining material is important to achieve a balance between comfort, heat transfer, and durability.

3.9 OUTER PROTECTIVE LAYER

The outer protective layer provides additional insulation and protects the glove from external environmental factors. It is typically made from durable materials such as synthetic fabrics, leather, or reinforced textile materials. This layer acts as a barrier against wind, moisture, dust, and physical damage, thereby increasing the lifespan of the glove. It also helps in retaining the heat within the glove by reducing heat dissipation to the surroundings. In addition to protection, the outer layer contributes to the structural strength and overall appearance of the glove. It ensures that the internal components, including the heat pack, remain safe during regular use. The design of this layer is important for achieving both durability and thermal efficiency.

3.10 ADVANTAGES OF THE SYSTEM

➤ Battery-Free Operation

No need for electricity or charging, making it reliable in all conditions.

➤ Reusable Design

Sodium acetate can be reheated and reused multiple times, reducing waste.

➤ Cost-Effective

Uses simple and inexpensive materials compared to electronic heated gloves.

➤ Safe and Non-Toxic

Sodium acetate is safe for skin contact and does not produce harmful substances.

➤ Eco-Friendly

Eliminates disposable waste and supports sustainable usage.

➤ Easy to Use

Simple snap-button activation without any technical complexity.

➤ Portable and Lightweight

Convenient to carry and comfortable to wear.

➤ No Maintenance Required

Does not involve circuits, batteries, or complex components.

➤ Provides Instant Heat

Heat is generated immediately after activation.

3.11 SUMMARY OF THE PROJECT

The proposed project focuses on the design and development of a battery-free heated glove system using sodium acetate as a heat-generating material. The system works on the principle of exothermic crystallization, where a supercooled sodium acetate solution releases heat when triggered by a snap button mechanism. This heat

is effectively transferred through the glove layers to provide warmth to the user's hands. The design integrates a sealed sodium acetate heat pack within a multi-layer glove structure to ensure efficient heat distribution, user comfort, and safety. Unlike conventional solutions, the proposed system does not require electrical power, making it simple, cost-effective, and reliable. Additionally, the heat pack can be regenerated by reheating, allowing multiple reuse cycles and reducing environmental impact.

Overall, the project provides an innovative and practical solution for cold protection by combining chemical heat generation, wearable design, and sustainability, making it suitable for everyday use in cold environments and potential therapeutic applications. In addition to its core functionality, the proposed system emphasizes user convenience and adaptability. The glove can be designed in various sizes and styles to suit different user groups, ensuring a comfortable fit and effective heat transfer. The lightweight nature of the system makes it easy to carry and use for extended periods without causing fatigue. The use of Sodium Acetate further enhances the sustainability aspect of the project, as it allows repeated usage without generating significant waste. This makes the system an environmentally responsible alternative to disposable heat packs and battery-powered heating devices.

Moreover, the project opens opportunities for future enhancements such as incorporating improved insulation materials, optimizing heat retention time, and refining the activation mechanism for better responsiveness. The design can also be extended to other wearable products like jackets, knee warmers, or therapeutic wraps, increasing its scope of application. From a practical perspective, the system is highly beneficial in scenarios such as outdoor work, travel, sports activities, and medical use where maintaining hand warmth is essential. Its simplicity, safety, and reusability make it a promising solution for real-world implementation.

Furthermore, the proposed system highlights the importance of low-cost innovation in developing practical solutions for everyday problems. The materials used in the glove and heat pack are easily available and affordable, making the product accessible to a wide range of users. This affordability, combined with reusability, significantly reduces long-term costs compared to disposable heating

alternatives. The design also ensures ease of maintenance, as the heat pack requires only simple reheating in hot water to restore its original state. This eliminates the need for specialized equipment or technical knowledge, making the system user-friendly. Additionally, the absence of electronic components minimizes the chances of failure, thereby increasing the durability and lifespan of the product.

In addition, the proposed heated glove system demonstrates strong potential for commercialization and large-scale production due to its simple design and minimal component requirements. The manufacturing process can be easily standardized, ensuring consistent quality and performance across units. With increasing demand for sustainable and energy-efficient wearable solutions, this product can cater to both consumer and healthcare markets. Its adaptability also allows customization based on climate conditions and user preferences, making it a versatile solution. Overall, the project not only addresses immediate thermal comfort needs but also contributes to the advancement of eco-friendly wearable technologies.

CHAPTER 4

RESULTS AND OBSERVATION

The developed sodium acetate-based hand warmer glove system was experimentally tested to evaluate its performance, efficiency, usability, and reliability under different operating conditions. The primary objective of the testing process was to verify whether the system could provide effective heating without the use of electrical power and to analyze its suitability as a practical solution for cold environments.

During the initial stage of observation, the sodium acetate solution was found to remain in a stable supercooled liquid state under normal room temperature conditions. This confirmed that the preparation and storage of the solution were carried out correctly. The stability of the supercooled state is important, as it ensures that the heat is stored in the form of latent energy until activation is required. No premature crystallization was observed, indicating good system reliability.

Upon activation, the snap metal button was pressed manually, which resulted in an immediate crystallization reaction. The transition from liquid to solid phase occurred rapidly, accompanied by a visible change in the texture of the solution. This confirmed that the nucleation process was successfully initiated by the mechanical disturbance created by the snap disc. The response time of the system was observed to be very short, typically within a few seconds, making it highly effective for instant heat generation.

The temperature rise of the heat pack was recorded qualitatively during the experiment. It was observed that the temperature increased to a comfortable level suitable for human use. The heat generated was sufficient to warm the hands without causing any discomfort or risk of burns. The maximum temperature achieved was within the safe operating range, demonstrating that the system is inherently safe due to the controlled nature of the chemical reaction. Unlike electrical heating systems, there is no risk of overheating or sudden temperature spikes.

The heat retention capability of the system was another important parameter observed during testing. The generated heat was maintained for a duration ranging between 30 to 60 minutes, depending on factors such as ambient temperature, glove insulation, and pack size. The temperature gradually decreased over time as heat was transferred to the surroundings. The rate of heat loss was found to be slower when better insulating materials were used in the glove design. This highlights the importance of proper material selection for improving system efficiency.

The role of the glove structure was analyzed in detail. The inner lining fabric played a significant role in ensuring uniform heat distribution across the hand. It prevented localized concentration of heat, thereby improving user comfort. The middle layer ensured proper placement of the heat pack, maintaining consistent contact with the hand surface. The outer insulation layer effectively reduced heat loss due to convection and radiation, thereby increasing the duration of warmth. The combination of these layers contributed to the overall performance of the system.

The usability of the system was also evaluated during the testing phase. It was observed that the snap button mechanism is simple and easy to operate. Users can activate the system instantly without requiring any technical knowledge or additional tools. This makes the design suitable for people of all age groups. The lightweight nature of the glove further enhances its usability, as it does not cause discomfort during prolonged use.

One of the major advantages observed during testing is the independence from electrical power. Unlike battery-operated heated gloves, the proposed system does not require charging or electrical connections. This eliminates issues such as battery failure, reduced efficiency in low temperatures, and maintenance requirements. As a result, the system is more reliable and suitable for outdoor and remote applications.

The reusability of the system was tested through multiple heating cycles. After the completion of the heat generation process, the sodium acetate solidified completely. The heat pack was then immersed in hot water until the crystals dissolved and returned to

liquid form. After cooling, the solution regained its supercooled state and was ready for reuse. This process was repeated several times, and the system maintained consistent performance without any degradation. This confirms the durability and sustainability of the design.

In addition to performance evaluation, safety aspects were carefully observed. No leakage was detected from the plastic pouch during repeated usage, indicating that the sealing was effective. The temperature remained within safe limits, and no harmful reactions were observed. Since sodium acetate is non-toxic, the system is safe for direct use in wearable applications. The absence of electrical components further reduces the risk of hazards such as short circuits or electrical shocks.

A comparative observation was also made between the proposed system and then the existing solutions. Compared to traditional insulated gloves, the proposed system provides active heat generation rather than just heat retention. When compared to battery-powered heated gloves, it offers a simpler and more cost-effective solution without the need for charging. In comparison with disposable chemical warmers, the proposed system is reusable and environmentally friendly, reducing waste generation.

However, certain limitations were identified during the observation process. The duration of heat generation is limited compared to electrical systems that can provide continuous heating. Additionally, the need for reheating the sodium acetate solution for reuse may not always be convenient in outdoor conditions. The heat output is also dependent on environmental conditions, which may affect performance in extremely low temperatures. Despite these limitations, the advantages of the system outweigh its drawbacks.

Further analysis indicates that the performance of the system can be improved by optimizing factors such as the size of the heat pack, concentration of the sodium acetate solution, and insulation quality of the glove. Increasing the thickness of insulation layers can reduce heat loss, while optimizing pack placement can improve heat distribution. These improvements can enhance the efficiency and usability of the system. From an application perspective, the system can be used in various scenarios

such as cold weather conditions, outdoor activities, medical therapy for improving blood circulation, and for individuals with sensitivity to cold. The simplicity and reliability of the design make it suitable for widespread adoption.

Overall, the results and observations clearly demonstrate that the proposed system sodium acetate-based hand warmer glove system is effective in providing a safe, reusable, and battery-free heating solution. The system successfully meets the design objectives and performs efficiently under normal conditions. The observations confirm that the integration of chemical heat generation with wearable design is a practical and innovative approach for cold protection.



Figure 4.1 Cozy Hands

This experiment highlights the practical application of phase change materials in everyday life, particularly in reusable heating devices. It also demonstrates the concept of supercooling and controlled energy release, which are important principles in thermal energy storage systems. The process is reversible, as the solidified sodium acetate can be reheated to return it to the liquid state, making the hand warmer reusable and environmentally friendly. The rapid and uniform spread of crystallization throughout the pack shows how efficiently the process occurs once initiated.

Additionally, the experiment highlights the reversible nature of the system, as the solidified pack can be reheated in hot water to dissolve the crystals and return the substance to its liquid state, ready for reuse.

Table 4.1. Temperature vs Time Observation

Time (minutes)	Temperature (°C)	Observation
0 (Before Activation)	25°C	Sodium acetate in liquid (supercooled) state
1	45°C	Snap button pressed, rapid heat generation begins
5	52°C	Maximum temperature reached, strong heat felt
10	50°C	Stable heat maintained
20	47°C	Gradual decrease in temperature
30	43°C	Warm condition continues
45	38°C	Heat reducing slowly
60	32°C	Mild warmth remains
70	28°C	Heat almost dissipated, solution solidified
Time (minutes)	Temperature (°C)	Observation
0 (Before Activation)	25°C	Sodium acetate in liquid (supercooled) state
1	45°C	Snap button pressed, rapid heat generation begins

This table depicts the performance of the sodium acetate heat pack based on temperature variation over time. At the initial stage (0 minutes), the solution is in a supercooled liquid state at 25°C. Once the snap button is pressed at 1 minute, rapid crystallization occurs, leading to a sudden increase in temperature to 45°C. The temperature reaches its maximum value of 52°C at 5 minutes, indicating peak heat generation. After this, the system maintains a relatively stable temperature around 50°C for a short duration. As time progresses, the temperature gradually decreases, reaching 47°C at 20 minutes and 43°C at 30 minutes while still providing warmth. Further cooling occurs over time, with the temperature dropping to 32°C at 60

minutes. By 70 minutes, the temperature reduces to 28°C, showing that the heat has almost dissipated and the solution has fully solidified.

Table 4.2 Performance Observation of the System

PARAMETER	OBSERVATION
Activation Time	Instant (within few seconds)
Maximum Temperature	~50–55°C
Heat Duration	30–60 minutes
Reusability	Can be reused multiple times
Safety	Safe and non-toxic
Power Requirement	No external power required
Ease of Use	Simple snap activation
Portability	Lightweight and easy to carry

The above table says the overall performance of the system based on various operational parameters. The activation time is observed to be instant, occurring within a few seconds after triggering. The system is capable of reaching a maximum temperature in the range of approximately 50–55°C, indicating effective heat generation. It provides a heat duration of about 30 to 60 minutes, ensuring sustained warmth for practical use. The system is reusable, allowing it to be used multiple times without degradation in performance. In terms of safety, it is non-toxic and safe for regular handling. Additionally, it does not require any external power source, making it energy-efficient and convenient. The ease of use is enhanced by a simple snap activation mechanism, and its lightweight nature ensures high portability for everyday use.

Table 4.3. Comparison with Existing Methods

Feature	Proposed System	Battery Heated Gloves	Disposable Warmers
Power Source	No battery	Battery required	No battery
Reusability	Reusable	Reusable	Single-use
Cost	Low	High	Medium
Environmental Impact	Eco-friendly	Moderate	High waste
Ease of Use	Simple	Moderate	Easy
Heat Duration	Moderate	Long	Moderate

The above table depicts a comparison between the proposed system, battery heated gloves, and disposable warmers based on key performance features. The proposed system operates without any battery, unlike battery heated gloves which require an external power source, while disposable warmers also do not need batteries. In terms of reusability, both the proposed system and battery heated gloves can be reused multiple times, whereas disposable warmers are designed for single use only. The cost of the proposed system is relatively low compared to the high cost of battery heated gloves and the moderate cost of disposable warmers. From an environmental perspective, the proposed system is eco-friendly, while battery heated gloves have a moderate impact and disposable warmers generate higher waste. Considering ease of use, the proposed system is simple to operate, battery heated gloves are moderately convenient, and disposable warmers are easy to use. Finally, the heat duration of the proposed system is moderate, similar to disposable warmers, while battery heated gloves provide longer heating duration.

CHAPTER 5

CONCLUSION AND FUTURE ENHANCEMENT

5.1 CONCLUSION

The development of the sodium acetate-based hand warmer glove system represents an effective and innovative approach to addressing the challenges associated with cold exposure. The primary objective of the project was to design a safe, reusable, and battery-free heating solution that can provide sufficient warmth to the user's hands. Based on the experimental results and observations, it can be concluded that the proposed system successfully achieves these objectives.

The system operates on the principle of exothermic crystallization, where the supercooled sodium acetate solution releases latent heat when triggered by a mechanical snap button. The results demonstrate that the activation process is instantaneous and reliable, providing quick heat generation within a few seconds. The temperature achieved is within a comfortable and safe range, ensuring that the user experiences effective warming without any risk of burns or discomfort.

The integration of the sodium acetate heat pack within a multi-layer glove structure significantly enhances the overall performance of the system. The inner lining ensures uniform heat distribution across the hand, while the outer insulation layer reduces heat loss and improves heat retention. This layered design contributes to both efficiency and user comfort. The glove remains lightweight, flexible, and easy to use, making it suitable for practical applications.

Another important conclusion is the system's reusability and sustainability. Unlike disposable hand warmers, the sodium acetate solution can be regenerated by simple reheating, allowing multiple cycles of use without degradation in performance. This reduces operational cost and minimizes environmental impact. Furthermore, the

absence of electrical components eliminates issues such as battery dependency, charging requirements, and electrical hazards, making the system more reliable and user-friendly.

The experimental observations also confirm that the system performs consistently under normal conditions. The heat generation duration, although limited, is sufficient for short-term exposure to cold environments. The system provides a practical balance between simplicity, safety, and performance. Overall, the project demonstrates that chemical-based heating systems can be effectively integrated into wearable designs to create efficient and sustainable thermal solutions.

5.2 FUTURE ENHANCEMENT

Although the proposed system shows promising results, there are several areas where further improvements can be made to enhance its performance, functionality, and applicability. One of the key areas for enhancement is increasing the duration of heat generation. This can be achieved by optimizing the concentration and volume of the sodium acetate solution or by using multiple heat packs within the glove. By increasing the thermal storage capacity, the system can provide longer-lasting warmth, making it more suitable for extended use in extremely cold environments.

Another important improvement involves the use of advanced insulation materials. Incorporating high-performance thermal fabrics or multi-layer insulation techniques can significantly reduce heat loss and improve overall efficiency. This would help in maintaining a stable temperature for a longer duration and enhance user comfort. The design of the glove can also be further improved for better ergonomics and functionality. For example, multiple smaller heat packs can be distributed across different sections of the glove, including the fingers, to ensure uniform heating. This would provide more effective coverage and improve the overall user experience.

In future developments, the system can be integrated with smart technologies to enhance control and monitoring. The addition of temperature sensors and indicators can help users monitor the heat level and ensure safe operation. Although the current system is completely mechanical, combining it with simple electronic monitoring systems can

improve usability without compromising safety. Another potential enhancement is the development of hybrid systems that combine chemical heating with minimal electrical support. Such systems can provide both instant heat generation and controlled temperature regulation, offering the advantages of both approaches.

The application of this technology can also be extended beyond gloves. Similar heating systems can be integrated into other wearable products such as jackets, socks, and therapeutic wraps. In the medical field, the system can be used for physiotherapy, muscle relaxation, and improving blood circulation in patients with certain health conditions. From a manufacturing perspective, improvements can be made in material quality, durability, and large-scale production techniques. Developing cost-effective manufacturing methods can make the product more accessible to a wider population. Finally, future research can focus on improving the efficiency of phase change materials and exploring alternative materials with higher heat storage capacity. This can further enhance the performance of the system and open new possibilities for thermal management applications.

Another important direction for future work involves improving thermal insulation. The efficiency of the system largely depends on how well the generated heat is retained within the glove. Advanced insulating materials such as aerogels, vacuum-insulated fabrics, or multi-layer composite textiles can be explored to significantly reduce heat loss. These materials offer superior thermal resistance while maintaining flexibility and comfort. By incorporating such materials, the system can achieve higher efficiency and longer heat retention without increasing the size or weight of the glove.

The design of the glove can also be enhanced to improve heat distribution. In the current system, the heat pack is typically placed in a single location, such as the palm or back of the hand. However, future designs can incorporate multiple smaller heat packs distributed across different regions, including fingers and joints. This would ensure uniform heating throughout the hand, improving comfort and effectiveness. Additionally, ergonomic improvements can be made to ensure better fit, flexibility, and user convenience. Customizable glove sizes and adjustable compartments for heat packs can further enhance usability.

Another promising area of development is the integration of smart technologies. Although the current system operates purely on a mechanical basis, future versions can include temperature sensors, indicators, or microcontrollers to monitor and regulate heat output. For example, a simple temperature indicator could inform the user about the current heat level, while advanced systems could automatically adjust heat distribution based on environmental conditions. Integration with wearable electronics and mobile applications could enable real-time monitoring and control, making the system more intelligent and user-friendly.

In addition to improving the glove design, future research can explore alternative phase change materials with higher thermal storage capacity. While sodium acetate is effective, other materials with higher latent heat values or longer heat retention properties may offer improved performance. Hybrid systems combining multiple phase change materials can also be developed to achieve staged heat release, providing a more consistent and prolonged heating effect.

The regeneration process of the system can also be optimized. Currently, reheating the sodium acetate solution requires access to hot water, which may not always be convenient in outdoor environments. Future designs can explore self-regenerating mechanisms or alternative methods such as solar heating or portable heating units. This would enhance the practicality of the system, especially in remote or off-grid locations.

From an application perspective, the technology can be extended beyond gloves to a wide range of wearable and non-wearable products. For instance, similar heating systems can be integrated into jackets, socks, shoes, and blankets to provide full-body warmth. In the healthcare sector, the system can be used for therapeutic purposes, such as improving blood circulation, relieving muscle stiffness, and providing localized heat therapy for patients with arthritis or other medical conditions. This opens up opportunities for the system to be used not only as a comfort device but also as a medical support tool.

Industrial and occupational applications also present significant opportunities. Workers in cold storage facilities, construction sites, and outdoor environments often face extreme temperatures that affect their productivity and safety. The proposed system can be adapted to provide reliable thermal protection in such conditions. Its simplicity, durability, and independence from electrical power make it particularly suitable for use in harsh environments.

Another important area for future enhancement is sustainability and environmental impact. While the current system is already more eco-friendly compared to disposable warmers, further improvements can be made by using biodegradable materials for the pouch and glove components. Additionally, optimizing the manufacturing process to reduce material waste and energy consumption can make the product more environmentally sustainable.

Large-scale production and commercialization of the system also present challenges and opportunities. Future work can focus on improving manufacturing techniques to ensure consistent quality and cost-effectiveness. Developing standardized designs and testing procedures can help in achieving mass production and market acceptance. Collaboration with textile industries and healthcare manufacturers can facilitate the commercialization process. Furthermore, research can be conducted to evaluate the long-term performance and durability of the system under different conditions. Testing the system in extreme environments, such as very low temperatures or high humidity, can provide valuable insights for further improvement.

Finally, interdisciplinary research combining material science, textile engineering, biomedical applications, and wearable technology can lead to the development of advanced thermal systems. The integration of innovative materials, smart sensors, and ergonomic designs can transform the proposed concept into a highly advanced and versatile product. With continuous improvement and research, the sodium acetate-based hand warmer system has the potential to evolve into a widely used solution for personal thermal management across various fields.

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